

APAC YouTube Trend Intelligence for Campaign ROI

NIKE APAC

SQL · Power BI · Python (Pandas, SciPy, ReportLab) — Regional Media Efficiency Analysis

Executive Summary

Campaign ROI has been highly volatile across our APAC marketing initiatives. Using SQL and Python, I extracted millions of trending video metrics from our regional databases, ran inferential statistical models, and built an analytics pipeline to track viewer behavior across borders. After identifying that the largest optimization opportunities lie in aligning campaign timing with regional attention velocity and adjusting target categories to cultural brand-safety thresholds, I recommend the following pivots:

- Restrict programmatic ad spend to the **Top 3 performing genres** per region to cut wasted budget on low-performing categories.
- Run **hyper-customized regional strategies** instead of a blanket APAC approach — statistical testing confirms audience friction differs meaningfully by market.
- Front-load spend in fast-decaying markets like **India** to capture the Day 1-2 viral surge.
- Channel premium storytelling budgets into **South Korea**, a high-engagement, low-friction market.

Business Problem

Ad spend efficiency is directly tied to campaign ROI and regional revenue growth. Stakeholders noticed blanket digital campaigns yielding volatile returns across Asian markets due to differing audience retention and brand-safety friction. **How can historical YouTube trending data tell us where, when, and in which genres audiences engage most — so budget decisions become data-driven?**

Regional Trending-Video Sample & Publishing-Hour Pulse (from view_cleaned_youtube_master)

INDIA 37,341 trending videos	SOUTH KOREA 1,161 trending videos	JAPAN 697 trending videos	PEAK HOUR (IN) 2 PM 7.6% share	PEAK HOUR (JP/KR) 9-10 AM ~10% share
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Methodology

- SQL query to extract, clean, and transform raw YouTube trending data.
- Power BI dashboard tracking regional metrics, velocity, and top genres.
- Python cross-regional distribution analysis (KDE, box/violin plots, heatmap) plus Kruskal-Wallis testing to isolate friction boundaries.

Skills

SQL: CTEs, Joins, Date/Time transforms, Group By, aggregates

Power BI: DAX, ETL, calculated columns, data-privacy filtering

Python: Pandas, Seaborn, SciPy (Kruskal-Wallis), pyodbc, custom functions

Key Statistical Findings

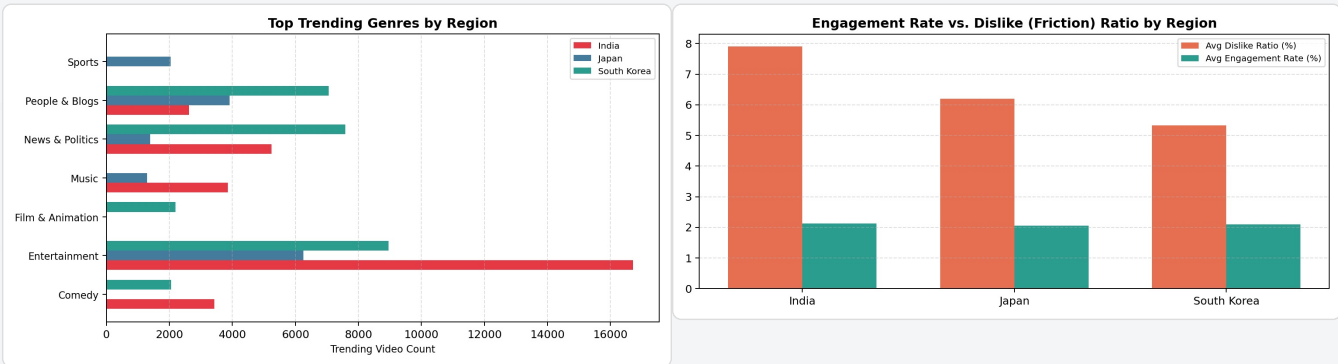
India dislike ratio: 7.9%

Japan: 6.2%

S. Korea: 5.3%

Kruskal-Wallis testing on **dislike/friction ratio** across regions was statistically significant ($H = 4886.9$, $p < 0.001$) — confirming real cultural differences in audience friction. Engagement rate differences across regions, however, were **not** statistically significant ($p = 0.667$), suggesting engagement itself is culturally uniform even though controversy/friction is not.

Regional Genre & Friction Visuals



Results & Business Recommendation

The Power BI dashboard gives marketing stakeholders self-serve visibility into regional audience dynamics, cutting ad hoc analytics requests. Entertainment, News & Politics, and People & Blogs dominate trending content in all three markets, while India shows the fastest viral velocity (peaking within 1-2 days of publish) versus a flatter curve in Japan and South Korea. Because friction (dislike ratio) — not raw engagement — is what varies meaningfully by culture, I recommend narrowing category footprint and aligning drop timing with each region's velocity curve and peak publishing window.

Next Steps

- A/B test localized creative across each region's top 3 genres to measure conversion variance.
- Train media-buying teams on the new Power BI dashboard.
- Monitor real-time delivery and retention during each region's peak publishing window (India: midday; Japan/S. Korea: 9-11 AM).